



US012402934B2

(12) **United States Patent**  
**Goodman et al.**

(10) **Patent No.:** **US 12,402,934 B2**  
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **ELECTROSURGICAL INSTRUMENT FOR GRASPING, TREATING, AND/OR DIVIDING TISSUE INCORPORATING THERMAL MANAGEMENT FEATURE**

(71) Applicant: **Covidien LP**, Mansfield, MA (US)

(72) Inventors: **Kelley D. Goodman**, Erie, CO (US);  
**Craig V. Krastins**, Arvada, CO (US);  
**Grant T. Sims**, Boulder, CO (US);  
**Daniel W. Mercier**, Erie, CO (US);  
**Jennifer L. Rich**, Parker, CO (US)

(73) Assignee: **Covidien LP**, Mansfield, MA (US)

(\*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 490 days.

(21) Appl. No.: **17/014,002**

(22) Filed: **Sep. 8, 2020**

(65) **Prior Publication Data**

US 2021/0077178 A1 Mar. 18, 2021

**Related U.S. Application Data**

(60) Provisional application No. 62/900,573, filed on Sep. 15, 2019.

(51) **Int. Cl.**  
**A61B 18/14** (2006.01)  
**A61B 18/00** (2006.01)

(52) **U.S. Cl.**  
CPC ..... **A61B 18/1442** (2013.01); **A61B 2018/00023** (2013.01); **A61B 2018/00077** (2013.01); **A61B 2018/00589** (2013.01); **A61B 2018/00595** (2013.01); **A61B 2018/0063** (2013.01)

(58) **Field of Classification Search**  
CPC ..... **A61B 2018/00023**; **A61B 2018/00077**;  
**A61B 18/1442**

See application file for complete search history.

(56) **References Cited**

**U.S. PATENT DOCUMENTS**

|             |         |                 |
|-------------|---------|-----------------|
| D249,549 S  | 9/1978  | Pike            |
| D263,020 S  | 2/1982  | Rau, III        |
| D295,893 S  | 5/1988  | Sharkany et al. |
| D295,894 S  | 5/1988  | Sharkany et al. |
| 4,763,669 A | 8/1988  | Jaeger          |
| D298,353 S  | 11/1988 | Manno           |

(Continued)

**FOREIGN PATENT DOCUMENTS**

|    |               |         |
|----|---------------|---------|
| EP | 0246350 A1    | 11/1987 |
| WO | 2012116957 A1 | 9/2012  |

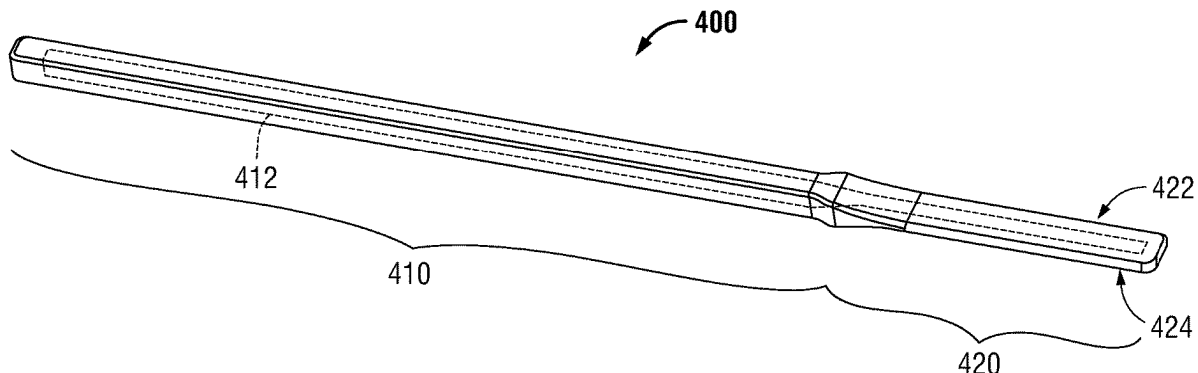
*Primary Examiner* — Joseph A Stoklosa

*Assistant Examiner* — Annie L Shoulders

(57) **ABSTRACT**

An electrosurgical instrument includes at least one shaft member, an end effector assembly extending distally from the at least one shaft member, and a heat pipe. The end effector assembly includes first and second jaw members, each including an electrically-conductive tissue-contacting surface. At least one of the first or second jaw members is movable relative to the other about a pivot between a spaced-apart position and an approximated position for grasping tissue between the electrically-conductive tissue-contacting surfaces. The heat pipe includes a proximal body portion and a flattened distal portion. The proximal body portion extends through at least a portion of the at least one shaft member, and the flattened distal portion extends through at least a portion of the first jaw member. The heat pipe is configured to facilitate drawing heat proximally from the end effector assembly.

**20 Claims, 6 Drawing Sheets**



(56)

## References Cited

## U.S. PATENT DOCUMENTS

|           |      |         |                     |           |      |         |                      |                        |
|-----------|------|---------|---------------------|-----------|------|---------|----------------------|------------------------|
| D299,413  | S    | 1/1989  | DeCarolis           | 7,641,653 | B2   | 1/2010  | Dalla Betta et al.   |                        |
| 5,100,420 | A    | 3/1992  | Green et al.        | 7,708,735 | B2   | 5/2010  | Chapman et al.       |                        |
| 5,258,001 | A    | 11/1993 | Corman              | D617,900  | S    | 6/2010  | Kingsley et al.      |                        |
| D343,453  | S    | 1/1994  | Noda                | D617,901  | S    | 6/2010  | Unger et al.         |                        |
| 5,304,203 | A    | 4/1994  | El-Mallawany et al. | D617,902  | S    | 6/2010  | Twomey et al.        |                        |
| D348,930  | S    | 7/1994  | Olson               | D617,903  | S    | 6/2010  | Unger et al.         |                        |
| D349,341  | S    | 8/1994  | Lichtman et al.     | D618,798  | S    | 6/2010  | Olson et al.         |                        |
| 5,344,424 | A    | 9/1994  | Roberts et al.      | D621,503  | S    | 8/2010  | Otten et al.         |                        |
| D354,564  | S    | 1/1995  | Medema              | D627,462  | S    | 11/2010 | Kingsley             |                        |
| D358,887  | S    | 5/1995  | Feinberg            | D628,289  | S    | 11/2010 | Romero               |                        |
| 5,540,685 | A    | 7/1996  | Parins et al.       | D628,290  | S    | 11/2010 | Romero               |                        |
| 5,578,052 | A    | 11/1996 | Koros et al.        | 7,854,185 | B2   | 12/2010 | Zhang et al.         |                        |
| 5,611,808 | A    | 3/1997  | Hossain et al.      | D630,324  | S    | 1/2011  | Reschke              |                        |
| 5,618,294 | A    | 4/1997  | Aust et al.         | 7,896,878 | B2   | 3/2011  | Johnson et al.       |                        |
| D384,413  | S    | 9/1997  | Zlock et al.        | D649,249  | S    | 11/2011 | Guerra               |                        |
| 5,665,100 | A    | 9/1997  | Yoon                | D649,643  | S    | 11/2011 | Allen, IV et al.     |                        |
| 5,752,644 | A    | 5/1998  | Bolanos et al.      | 8,100,894 | B2 * | 1/2012  | Mucko                | A61B 18/1442<br>606/51 |
| H1745     | H    | 8/1998  | Paraschac           | 8,147,489 | B2   | 4/2012  | Moses et al.         |                        |
| 5,814,043 | A    | 9/1998  | Shapeton            | D661,394  | S    | 6/2012  | Romero et al.        |                        |
| D402,028  | S    | 12/1998 | Grimm et al.        | 8,292,880 | B2 * | 10/2012 | Prakash              | A61B 18/1815<br>606/41 |
| D408,018  | S    | 4/1999  | McNaughton          | 8,298,233 | B2   | 10/2012 | Mueller              |                        |
| 5,913,874 | A    | 6/1999  | Berns et al.        | D670,808  | S    | 11/2012 | Moua et al.          |                        |
| 5,960,544 | A    | 10/1999 | Beyers              | 8,366,709 | B2   | 2/2013  | Schechter et al.     |                        |
| D416,089  | S    | 11/1999 | Barton et al.       | 8,394,096 | B2   | 3/2013  | Moses et al.         |                        |
| 6,050,996 | A    | 4/2000  | Schmaltz et al.     | D680,220  | S    | 4/2013  | Rachlin              |                        |
| D424,694  | S    | 5/2000  | Tetzlaff et al.     | 8,409,246 | B2   | 4/2013  | Kerr et al.          |                        |
| D425,201  | S    | 5/2000  | Tetzlaff et al.     | 8,409,247 | B2   | 4/2013  | Garrison et al.      |                        |
| H1904     | H    | 10/2000 | Yates et al.        | 8,425,504 | B2   | 4/2013  | Orton et al.         |                        |
| 6,293,954 | B1   | 9/2001  | Fogarty et al.      | 8,425,511 | B2   | 4/2013  | Olson                |                        |
| D449,886  | S    | 10/2001 | Tetzlaff et al.     | 8,430,877 | B2   | 4/2013  | Kerr et al.          |                        |
| 6,329,778 | B1   | 12/2001 | Culp et al.         | 8,439,913 | B2   | 5/2013  | Horner et al.        |                        |
| 6,334,861 | B1   | 1/2002  | Chandler et al.     | 8,469,716 | B2   | 6/2013  | Fedotov et al.       |                        |
| D453,923  | S    | 2/2002  | Olson               | 8,469,991 | B2   | 6/2013  | Kerr                 |                        |
| D454,951  | S    | 3/2002  | Bon                 | 8,469,992 | B2   | 6/2013  | Roy et al.           |                        |
| D457,958  | S    | 5/2002  | Dycus et al.        | 8,480,671 | B2   | 7/2013  | Mueller              |                        |
| D457,959  | S    | 5/2002  | Tetzlaff et al.     | 8,491,624 | B2   | 7/2013  | Kerr et al.          |                        |
| 6,406,485 | B1   | 6/2002  | Hossain et al.      | 8,491,625 | B2   | 7/2013  | Horner               |                        |
| H2037     | H    | 7/2002  | Yates et al.        | 8,491,626 | B2   | 7/2013  | Roy et al.           |                        |
| 6,464,704 | B2   | 10/2002 | Schmaltz et al.     | 8,512,336 | B2   | 8/2013  | Couture              |                        |
| D465,281  | S    | 11/2002 | Lang                | 8,540,749 | B2   | 9/2013  | Garrison et al.      |                        |
| D466,209  | S    | 11/2002 | Bon                 | 8,551,091 | B2   | 10/2013 | Couture et al.       |                        |
| 6,511,480 | B1   | 1/2003  | Tetzlaff et al.     | 8,556,929 | B2   | 10/2013 | Harper et al.        |                        |
| 6,673,092 | B1   | 1/2004  | Bacher              | 8,568,397 | B2   | 10/2013 | Horner et al.        |                        |
| D493,888  | S    | 8/2004  | Reschke             | 8,568,408 | B2   | 10/2013 | Townsend et al.      |                        |
| D496,997  | S    | 10/2004 | Dycus et al.        | 8,585,736 | B2   | 11/2013 | Horner et al.        |                        |
| 6,800,077 | B1 * | 10/2004 | Mucko               | 8,591,510 | B2   | 11/2013 | Allen, IV et al.     |                        |
|           |      |         |                     | 8,597,295 | B2   | 12/2013 | Kerr                 |                        |
| D499,181  | S    | 11/2004 | Dycus et al.        | 8,623,018 | B2   | 1/2014  | Horner et al.        |                        |
| D502,994  | S    | 3/2005  | Blake, III          | 8,628,557 | B2   | 1/2014  | Collings et al.      |                        |
| D509,297  | S    | 9/2005  | Wells               | 8,641,712 | B2   | 2/2014  | Couture              |                        |
| 6,942,662 | B2 * | 9/2005  | Goble               | 8,647,343 | B2   | 2/2014  | Chojin et al.        |                        |
|           |      |         |                     | 8,652,135 | B2   | 2/2014  | Nau, Jr.             |                        |
| D525,361  | S    | 7/2006  | Hushka              | 8,663,222 | B2   | 3/2014  | Anderson et al.      |                        |
| 7,103,947 | B2   | 9/2006  | Sartor et al.       | 8,672,939 | B2   | 3/2014  | Garrison             |                        |
| D531,311  | S    | 10/2006 | Guerra et al.       | 8,679,098 | B2   | 3/2014  | Hart                 |                        |
| 7,118,570 | B2   | 10/2006 | Tetzlaff et al.     | 8,685,009 | B2   | 4/2014  | Chernov et al.       |                        |
| D533,274  | S    | 12/2006 | Visconti et al.     | 8,685,021 | B2   | 4/2014  | Chernov et al.       |                        |
| D533,942  | S    | 12/2006 | Kerr et al.         | 8,685,056 | B2   | 4/2014  | Evans et al.         |                        |
| 7,147,638 | B2   | 12/2006 | Chapman et al.      | 8,702,737 | B2   | 4/2014  | Chojin et al.        |                        |
| D535,027  | S    | 1/2007  | James et al.        | 8,702,749 | B2   | 4/2014  | Twomey               |                        |
| 7,169,146 | B2   | 1/2007  | Truckai et al.      | 8,734,445 | B2   | 5/2014  | Johnson et al.       |                        |
| D538,932  | S    | 3/2007  | Malik               | 8,740,898 | B2   | 6/2014  | Chojin et al.        |                        |
| D541,418  | S    | 4/2007  | Schechter et al.    | 8,745,840 | B2   | 6/2014  | Hempstead et al.     |                        |
| D541,611  | S    | 5/2007  | Aglassing           | 8,747,434 | B2   | 6/2014  | Larson et al.        |                        |
| D541,938  | S    | 5/2007  | Kerr et al.         | 8,756,785 | B2   | 6/2014  | Allen, IV et al.     |                        |
| 7,220,951 | B2   | 5/2007  | Truckai et al.      | 8,784,418 | B2   | 7/2014  | Romero               |                        |
| D545,432  | S    | 6/2007  | Watanabe            | 8,795,269 | B2   | 8/2014  | Garrison             |                        |
| D547,154  | S    | 7/2007  | Lee                 | 8,808,288 | B2   | 8/2014  | Reschke              |                        |
| 7,329,257 | B2   | 2/2008  | Kanehira et al.     | 8,814,864 | B2   | 8/2014  | Gilbert              |                        |
| D564,662  | S    | 3/2008  | Moses et al.        | 8,840,639 | B2   | 9/2014  | Gerhardt, Jr. et al. |                        |
| D567,943  | S    | 4/2008  | Moses et al.        | 8,845,636 | B2   | 9/2014  | Allen, IV et al.     |                        |
| D575,395  | S    | 8/2008  | Hushka              | 8,852,185 | B2   | 10/2014 | Twomey               |                        |
| D575,401  | S    | 8/2008  | Hixson et al.       | 8,852,228 | B2   | 10/2014 | Nau, Jr.             |                        |
| 7,431,730 | B2   | 10/2008 | Viola               | 8,858,553 | B2   | 10/2014 | Chojin               |                        |
| D582,038  | S    | 12/2008 | Swoyer et al.       | 8,864,753 | B2   | 10/2014 | Nau, Jr. et al.      |                        |
|           |      |         |                     | 8,864,795 | B2   | 10/2014 | Kerr et al.          |                        |
|           |      |         |                     | 8,887,373 | B2   | 11/2014 | Brandt et al.        |                        |

(56)

## References Cited

## U.S. PATENT DOCUMENTS

|              |      |         |                    |              |    |         |                  |
|--------------|------|---------|--------------------|--------------|----|---------|------------------|
| 8,888,771    | B2   | 11/2014 | Twomey             | 2011/0054469 | A1 | 3/2011  | Kappus et al.    |
| 8,888,775    | B2   | 11/2014 | Nau, Jr. et al.    | 2011/0060314 | A1 | 3/2011  | Wallace et al.   |
| 8,898,888    | B2   | 12/2014 | Brandt et al.      | 2011/0060356 | A1 | 3/2011  | Reschke et al.   |
| 8,900,232    | B2   | 12/2014 | Ourada             | 2011/0072638 | A1 | 3/2011  | Brandt et al.    |
| 8,906,018    | B2   | 12/2014 | Rooks et al.       | 2011/0087218 | A1 | 4/2011  | Boudreaux et al. |
| 8,920,421    | B2   | 12/2014 | Rupp               | 2011/0218530 | A1 | 9/2011  | Reschke          |
| 8,932,293    | B2   | 1/2015  | Chernov et al.     | 2011/0238065 | A1 | 9/2011  | Hunt et al.      |
| 8,936,614    | B2   | 1/2015  | Allen, IV          | 2011/0238067 | A1 | 9/2011  | Moses et al.     |
| 8,939,972    | B2   | 1/2015  | Twomey             | 2011/0257680 | A1 | 10/2011 | Reschke et al.   |
| 8,945,175    | B2   | 2/2015  | Twomey             | 2011/0270245 | A1 | 11/2011 | Horner et al.    |
| 8,961,504    | B2   | 2/2015  | Hoarau et al.      | 2011/0270251 | A1 | 11/2011 | Horner et al.    |
| 8,968,283    | B2   | 3/2015  | Kharin             | 2011/0276049 | A1 | 11/2011 | Gerhardt         |
| 8,968,305    | B2   | 3/2015  | Dumbauld et al.    | 2011/0295313 | A1 | 12/2011 | Kerr             |
| 8,968,316    | B2   | 3/2015  | Roy et al.         | 2012/0059372 | A1 | 3/2012  | Johnson          |
| 8,968,357    | B2   | 3/2015  | Mueller            | 2012/0059409 | A1 | 3/2012  | Reschke et al.   |
| 8,968,359    | B2   | 3/2015  | Kerr et al.        | 2012/0083785 | A1 | 4/2012  | Roy et al.       |
| 9,005,200    | B2   | 4/2015  | Roy et al.         | 2012/0083786 | A1 | 4/2012  | Artale et al.    |
| 9,017,372    | B2   | 4/2015  | Artale et al.      | 2012/0083827 | A1 | 4/2012  | Artale et al.    |
| 9,028,484    | B2   | 5/2015  | Craig              | 2012/0123402 | A1 | 5/2012  | Chernov et al.   |
| 9,028,492    | B2   | 5/2015  | Kerr et al.        | 2012/0123404 | A1 | 5/2012  | Craig            |
| 9,028,495    | B2   | 5/2015  | Mueller et al.     | 2012/0123410 | A1 | 5/2012  | Craig            |
| 9,039,704    | B2   | 5/2015  | Joseph             | 2012/0130367 | A1 | 5/2012  | Garrison         |
| 9,039,732    | B2   | 5/2015  | Sims et al.        | 2012/0136354 | A1 | 5/2012  | Rupp             |
| 9,084,608    | B2   | 7/2015  | Larson et al.      | 2012/0172868 | A1 | 7/2012  | Twomey et al.    |
| 9,113,933    | B2   | 8/2015  | Chernova et al.    | 2012/0172873 | A1 | 7/2012  | Artale et al.    |
| 9,113,934    | B2   | 8/2015  | Chernov et al.     | 2012/0172924 | A1 | 7/2012  | Allen, IV        |
| 9,161,807    | B2   | 10/2015 | Garrison           | 2012/0184988 | A1 | 7/2012  | Twomey et al.    |
| 9,211,657    | B2   | 12/2015 | Ackley et al.      | 2012/0184989 | A1 | 7/2012  | Twomey           |
| 9,265,565    | B2   | 2/2016  | Kerr               | 2012/0184990 | A1 | 7/2012  | Twomey           |
| 9,265,568    | B2   | 2/2016  | Chernov et al.     | 2012/0209263 | A1 | 8/2012  | Sharp et al.     |
| 9,333,002    | B2   | 5/2016  | Garrison           | 2012/0215219 | A1 | 8/2012  | Roy et al.       |
| 9,381,059    | B2   | 7/2016  | Garrison           | 2012/0239034 | A1 | 9/2012  | Horner et al.    |
| 9,439,710    | B2   | 9/2016  | Reu et al.         | 2012/0253344 | A1 | 10/2012 | Dumbauld et al.  |
| 9,456,870    | B2   | 10/2016 | Chernov et al.     | 2012/0259331 | A1 | 10/2012 | Garrison         |
| 9,498,278    | B2   | 11/2016 | Couture et al.     | 2012/0265241 | A1 | 10/2012 | Hart et al.      |
| 9,498,279    | B2   | 11/2016 | Artale et al.      | 2012/0283727 | A1 | 11/2012 | Twomey           |
| 9,504,519    | B2   | 11/2016 | Kerr et al.        | 2012/0296205 | A1 | 11/2012 | Chernov et al.   |
| 9,585,709    | B2   | 3/2017  | Krapohl            | 2012/0296238 | A1 | 11/2012 | Chernov et al.   |
| 9,615,877    | B2   | 4/2017  | Tyrrell et al.     | 2012/0296239 | A1 | 11/2012 | Chernov et al.   |
| 9,655,672    | B2   | 5/2017  | Artale et al.      | 2012/0296317 | A1 | 11/2012 | Chernov et al.   |
| 9,962,221    | B2   | 5/2018  | Lee et al.         | 2012/0296323 | A1 | 11/2012 | Chernov et al.   |
| 9,987,078    | B2   | 6/2018  | Thomson et al.     | 2012/0296324 | A1 | 11/2012 | Chernov et al.   |
| 2002/0016591 | A1 * | 2/2002  | Levine             | 2012/0296334 | A1 | 11/2012 | Kharin           |
|              |      |         | A61B 18/1442       | 2012/0303025 | A1 | 11/2012 | Garrison         |
|              |      |         | 606/51             | 2012/0323238 | A1 | 12/2012 | Tyrrell et al.   |
|              |      |         |                    | 2012/0330308 | A1 | 12/2012 | Joseph           |
|              |      |         |                    | 2012/0330309 | A1 | 12/2012 | Joseph           |
| 2003/0018332 | A1   | 1/2003  | Schmaltz et al.    | 2013/0018364 | A1 | 1/2013  | Chernov et al.   |
| 2003/0109875 | A1   | 6/2003  | Tetzlaff et al.    | 2013/0018372 | A1 | 1/2013  | Sims et al.      |
| 2003/0199869 | A1   | 10/2003 | Johnson et al.     | 2013/0018411 | A1 | 1/2013  | Collings et al.  |
| 2003/0220637 | A1   | 11/2003 | Truckai et al.     | 2013/0022495 | A1 | 1/2013  | Allen, IV et al. |
| 2003/0229344 | A1   | 12/2003 | Dycus et al.       | 2013/0030432 | A1 | 1/2013  | Garrison et al.  |
| 2004/0092927 | A1   | 5/2004  | Podhajsky et al.   | 2013/0041370 | A1 | 2/2013  | Unger            |
| 2005/0070889 | A1   | 3/2005  | Nobis et al.       | 2013/0046295 | A1 | 2/2013  | Kerr et al.      |
| 2005/0107784 | A1   | 5/2005  | Moses et al.       | 2013/0046303 | A1 | 2/2013  | Evans et al.     |
| 2005/0113826 | A1   | 5/2005  | Johnson et al.     | 2013/0046306 | A1 | 2/2013  | Evans et al.     |
| 2005/0113828 | A1   | 5/2005  | Shields et al.     | 2013/0046337 | A1 | 2/2013  | Evans et al.     |
| 2005/0159745 | A1   | 7/2005  | Truckai et al.     | 2013/0060250 | A1 | 3/2013  | Twomey et al.    |
| 2006/0253126 | A1   | 11/2006 | Bjerken et al.     | 2013/0066318 | A1 | 3/2013  | Kerr             |
| 2007/0062017 | A1   | 3/2007  | Dycus et al.       | 2013/0071282 | A1 | 3/2013  | Fry              |
| 2007/0088356 | A1   | 4/2007  | Moses et al.       | 2013/0072927 | A1 | 3/2013  | Allen, IV et al. |
| 2007/0179499 | A1   | 8/2007  | Garrison           | 2013/0079760 | A1 | 3/2013  | Twomey et al.    |
| 2007/0260241 | A1   | 11/2007 | Dalla Betta et al. | 2013/0079762 | A1 | 3/2013  | Twomey et al.    |
| 2008/0215048 | A1   | 9/2008  | Hafner et al.      | 2013/0079774 | A1 | 3/2013  | Whitney et al.   |
| 2009/0131934 | A1   | 5/2009  | Odom et al.        | 2013/0085491 | A1 | 4/2013  | Twomey et al.    |
| 2009/0171353 | A1   | 7/2009  | Johnson et al.     | 2013/0085496 | A1 | 4/2013  | Unger et al.     |
| 2009/0182327 | A1   | 7/2009  | Unger              | 2013/0103030 | A1 | 4/2013  | Garrison         |
| 2009/0240246 | A1   | 9/2009  | Deville et al.     | 2013/0103031 | A1 | 4/2013  | Garrison         |
| 2009/0302090 | A1   | 12/2009 | Shah               | 2013/0103035 | A1 | 4/2013  | Horner et al.    |
| 2009/0308909 | A1   | 12/2009 | Nalagatla et al.   | 2013/0123837 | A1 | 5/2013  | Roy et al.       |
| 2010/0016857 | A1   | 1/2010  | McKenna et al.     | 2013/0138101 | A1 | 5/2013  | Kerr             |
| 2010/0130977 | A1   | 5/2010  | Garrison et al.    | 2013/0138102 | A1 | 5/2013  | Twomey et al.    |
| 2010/0179545 | A1   | 7/2010  | Twomey et al.      | 2013/0138129 | A1 | 5/2013  | Garrison et al.  |
| 2010/0179547 | A1   | 7/2010  | Cunningham et al.  | 2013/0144284 | A1 | 6/2013  | Behnke et al.    |
| 2010/0228250 | A1   | 9/2010  | Brogna             | 2013/0178852 | A1 | 7/2013  | Allen, IV et al. |
| 2010/0274244 | A1   | 10/2010 | Heard              | 2013/0185922 | A1 | 7/2013  | Twomey et al.    |
| 2010/0292691 | A1   | 11/2010 | Brogna             | 2013/0190753 | A1 | 7/2013  | Garrison et al.  |
| 2010/0305567 | A1   | 12/2010 | Swanson            | 2013/0190760 | A1 | 7/2013  | Allen, IV et al. |
|              |      |         |                    | 2013/0197503 | A1 | 8/2013  | Orszulak         |

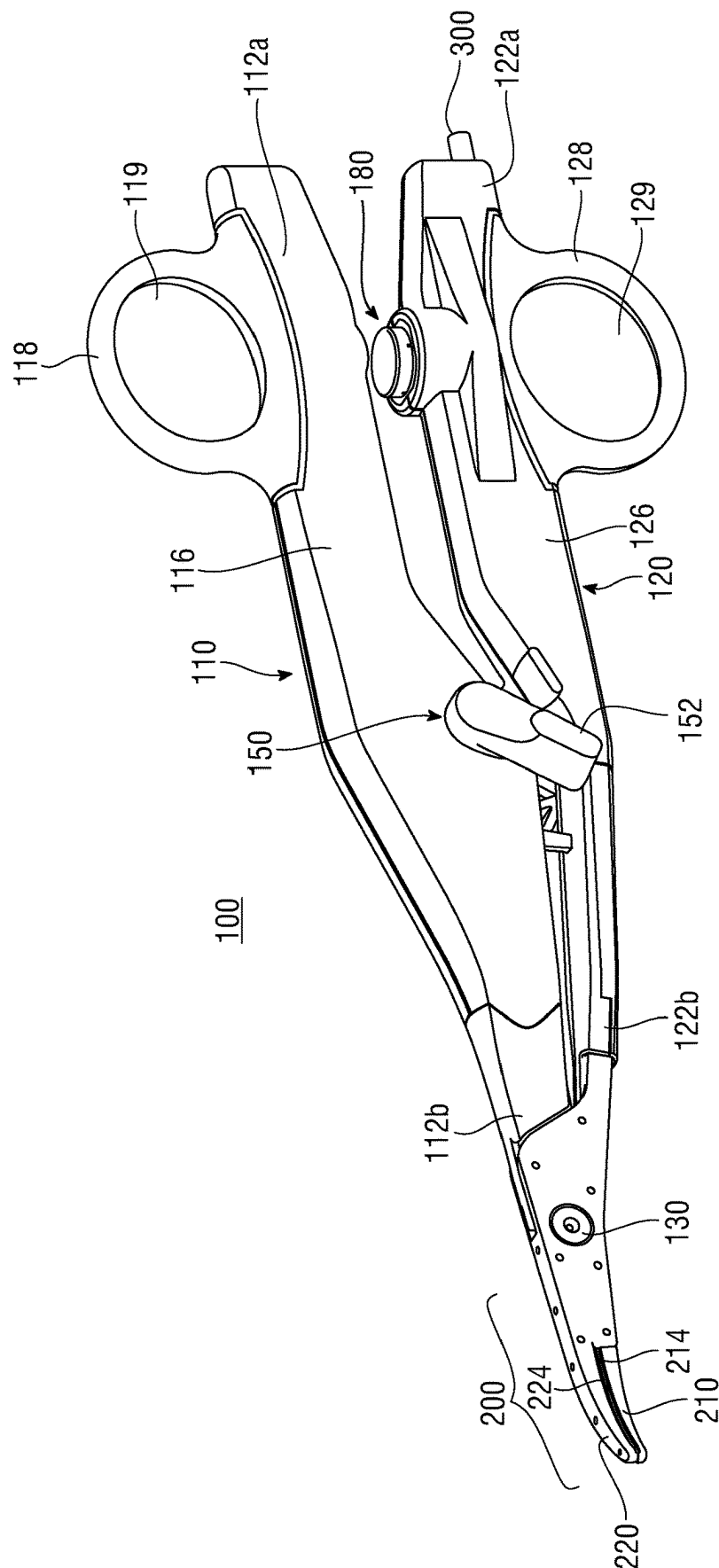
(56)

**References Cited**

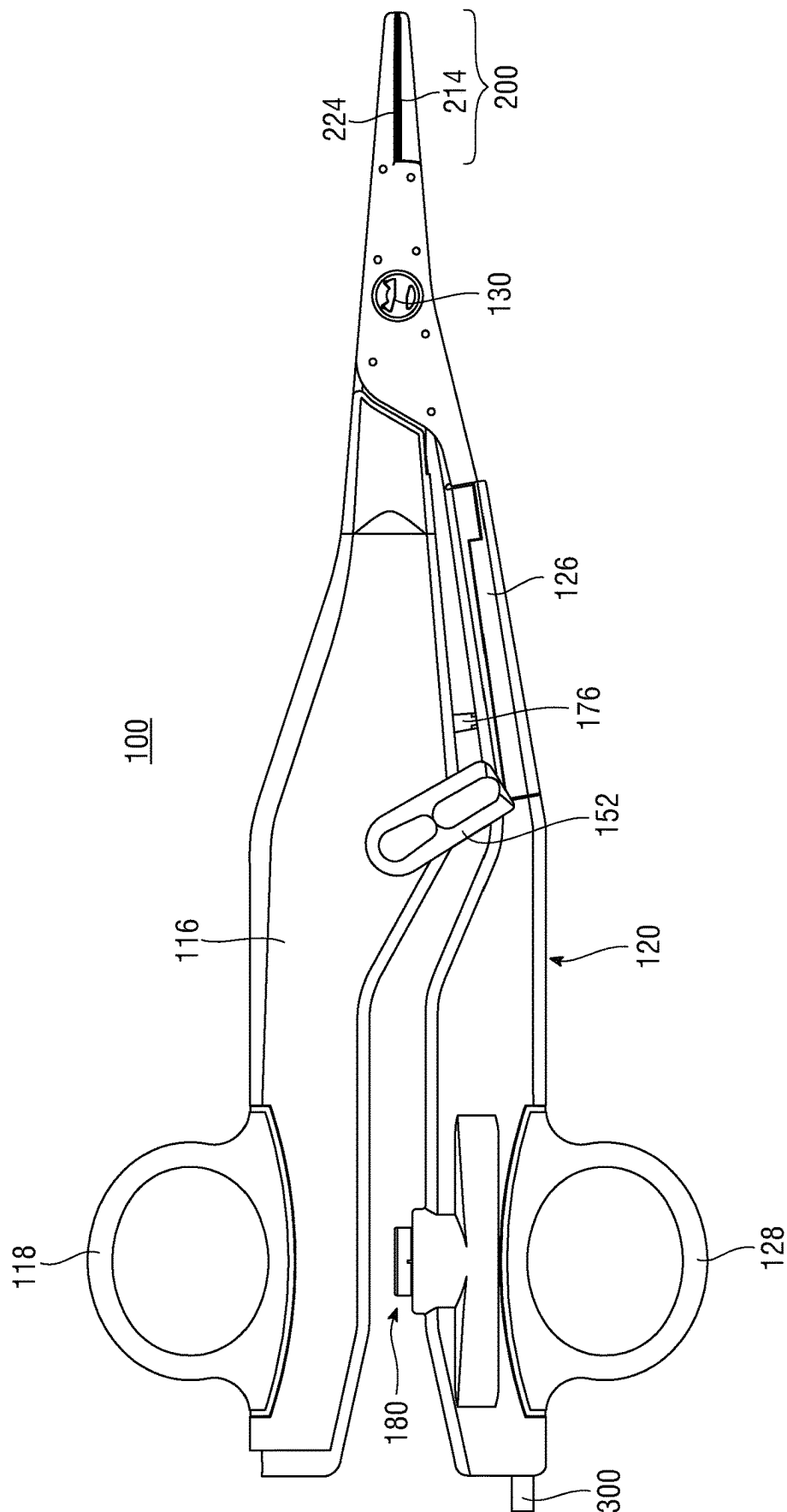
## U.S. PATENT DOCUMENTS

|              |    |        |                        |              |     |         |                               |
|--------------|----|--------|------------------------|--------------|-----|---------|-------------------------------|
| 2013/0226177 | A1 | 8/2013 | Brandt et al.          | 2014/0330308 | A1  | 11/2014 | Hart et al.                   |
| 2014/0221994 | A1 | 8/2014 | Reschke                | 2014/0336635 | A1  | 11/2014 | Hart et al.                   |
| 2014/0221995 | A1 | 8/2014 | Guerra et al.          | 2014/0353188 | A1  | 12/2014 | Reschke et al.                |
| 2014/0221999 | A1 | 8/2014 | Cunningham et al.      | 2015/0018816 | A1  | 1/2015  | Latimer                       |
| 2014/0228842 | A1 | 8/2014 | Dycus et al.           | 2015/0025528 | A1  | 1/2015  | Arts                          |
| 2014/0230243 | A1 | 8/2014 | Roy et al.             | 2015/0032106 | A1  | 1/2015  | Rachlin                       |
| 2014/0236149 | A1 | 8/2014 | Kharin et al.          | 2015/0051598 | A1  | 2/2015  | Orszulak et al.               |
| 2014/0243811 | A1 | 8/2014 | Reschke et al.         | 2015/0051640 | A1  | 2/2015  | Twomey et al.                 |
| 2014/0243824 | A1 | 8/2014 | Gilbert                | 2015/0066026 | A1  | 3/2015  | Hart et al.                   |
| 2014/0249528 | A1 | 9/2014 | Hixson et al.          | 2015/0066076 | A1  | 3/2015  | Kerr et al.                   |
| 2014/0250686 | A1 | 9/2014 | Hempstead et al.       | 2015/0080889 | A1  | 3/2015  | Cunningham et al.             |
| 2014/0257274 | A1 | 9/2014 | McCullough, Jr. et al. | 2015/0082928 | A1  | 3/2015  | Kappus et al.                 |
| 2014/0257283 | A1 | 9/2014 | Johnson et al.         | 2015/0088122 | A1  | 3/2015  | Jensen                        |
| 2014/0257284 | A1 | 9/2014 | Artale                 | 2015/0088126 | A1  | 3/2015  | Duffin et al.                 |
| 2014/0257285 | A1 | 9/2014 | Moua                   | 2015/0088128 | A1  | 3/2015  | Couture                       |
| 2014/0276803 | A1 | 9/2014 | Hart                   | 2015/0094714 | A1  | 4/2015  | Lee et al.                    |
| 2014/0284313 | A1 | 9/2014 | Allen, IV et al.       | 2016/0157925 | A1  | 6/2016  | Artale et al.                 |
| 2014/0288549 | A1 | 9/2014 | McKenna et al.         | 2016/0175031 | A1  | 6/2016  | Boudreaux                     |
| 2014/0288553 | A1 | 9/2014 | Johnson et al.         | 2017/0128120 | A1  | 5/2017  | Cho et al.                    |
|              |    |        |                        | 2018/0199986 | A1* | 7/2018  | Ding ..... A61B 18/1445       |
|              |    |        |                        | 2018/0353162 | A1  | 12/2018 | Hancock et al.                |
|              |    |        |                        | 2018/0368910 | A1* | 12/2018 | Kirwan, Jr. .... A61B 18/1445 |

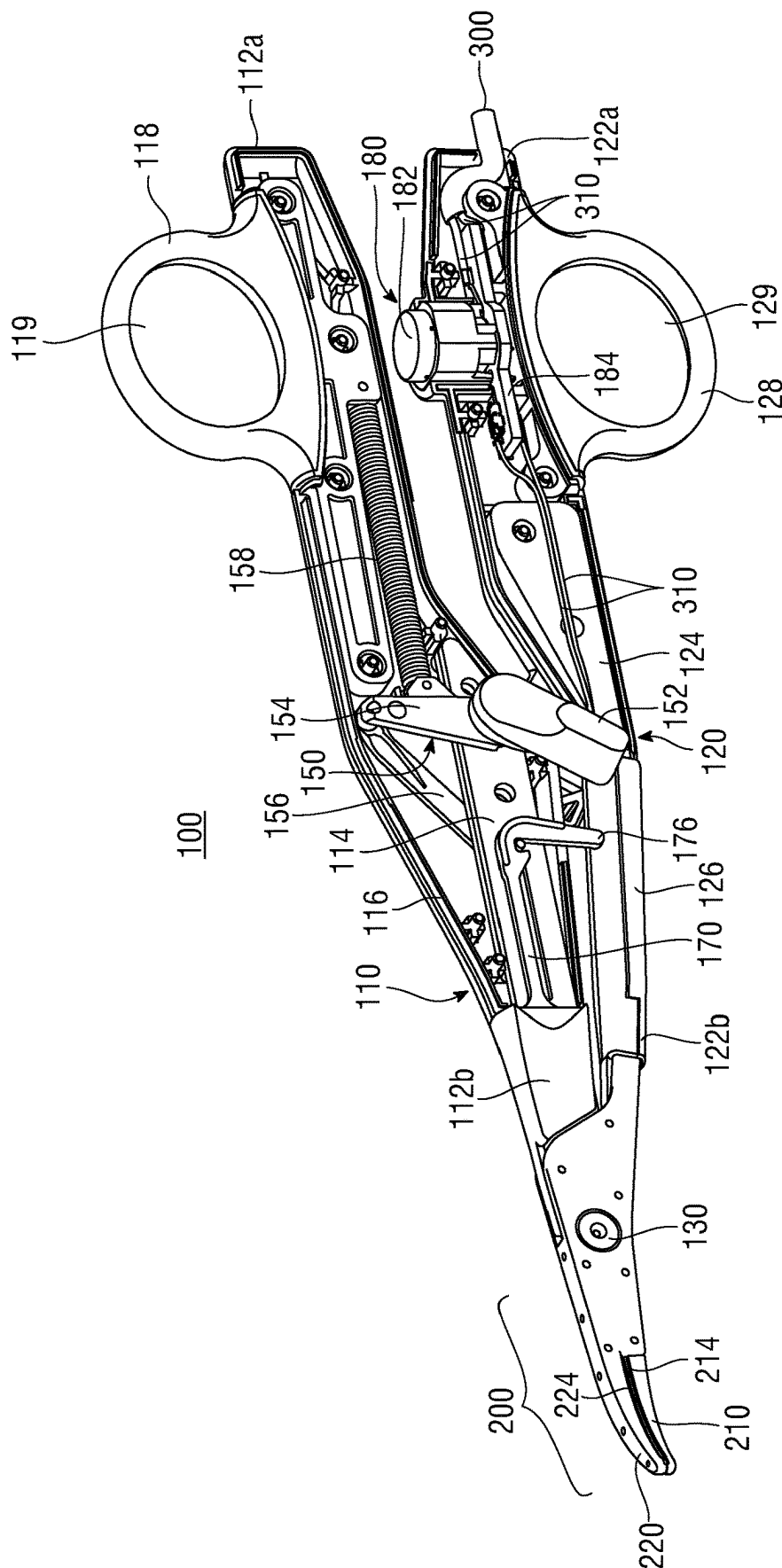
\* cited by examiner



**FIG. 1A**



**FIG. 1B**



**FIG. 2A**

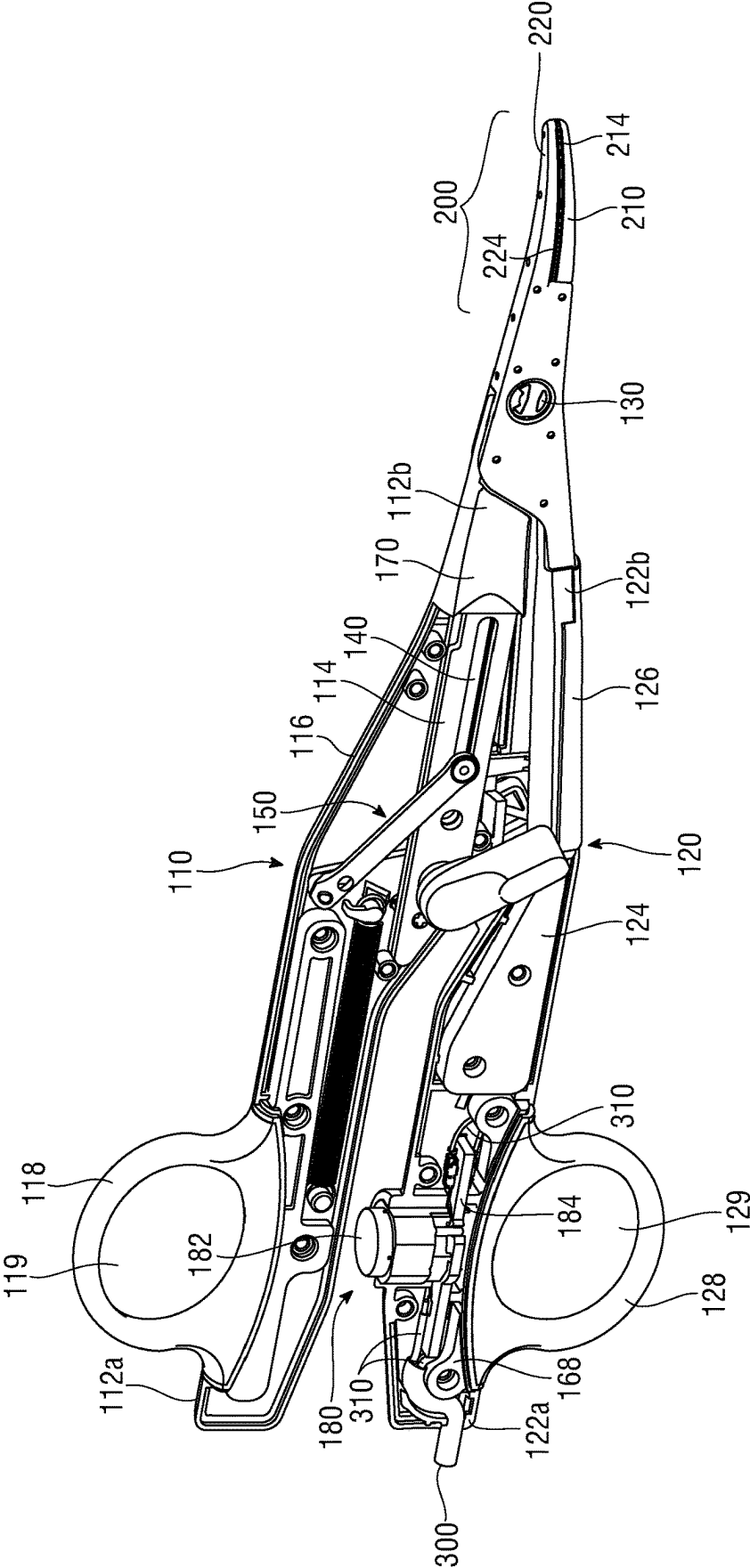
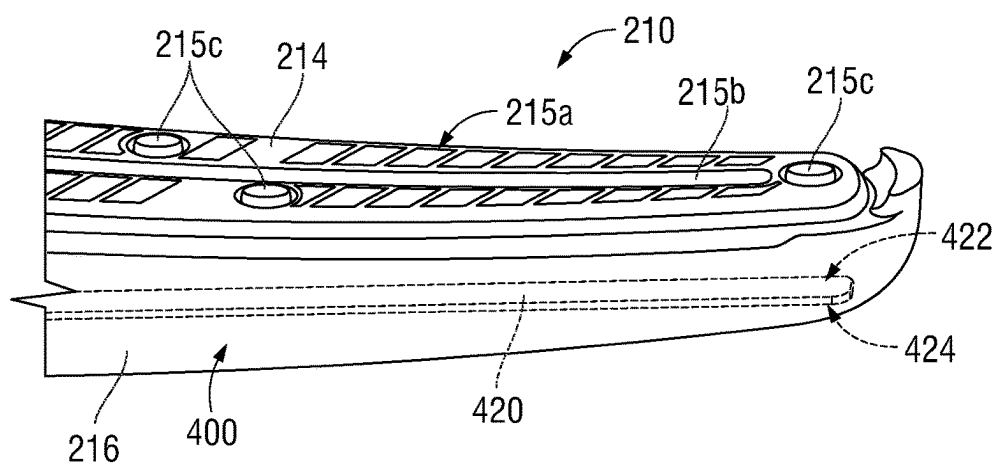
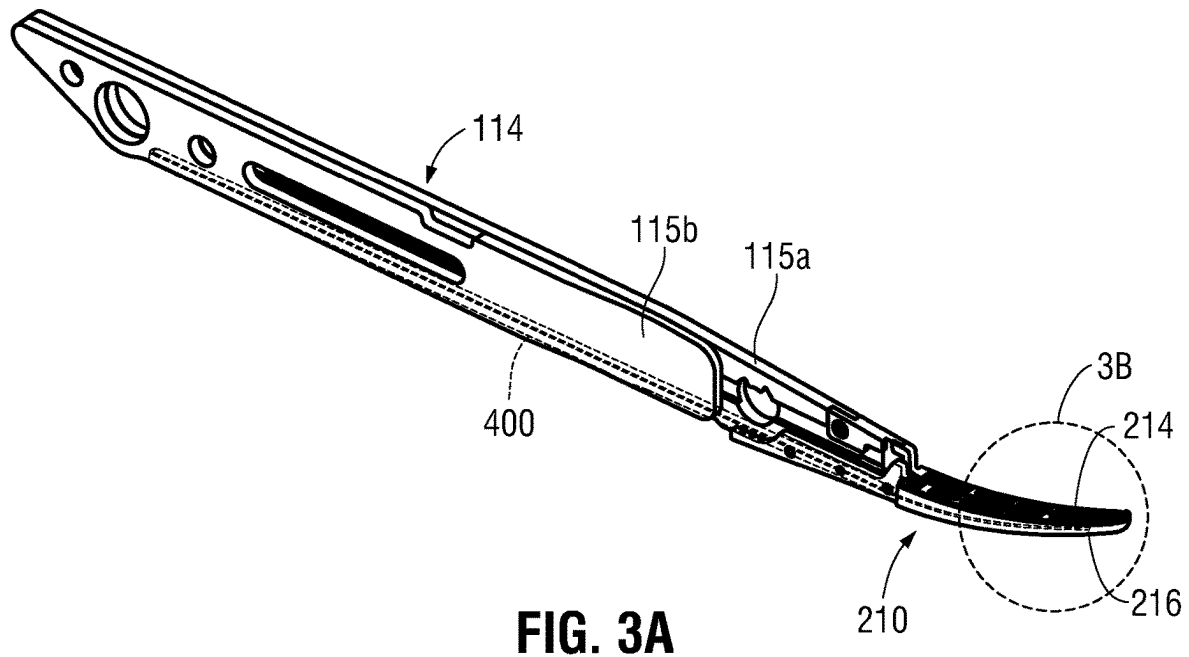


FIG. 2B





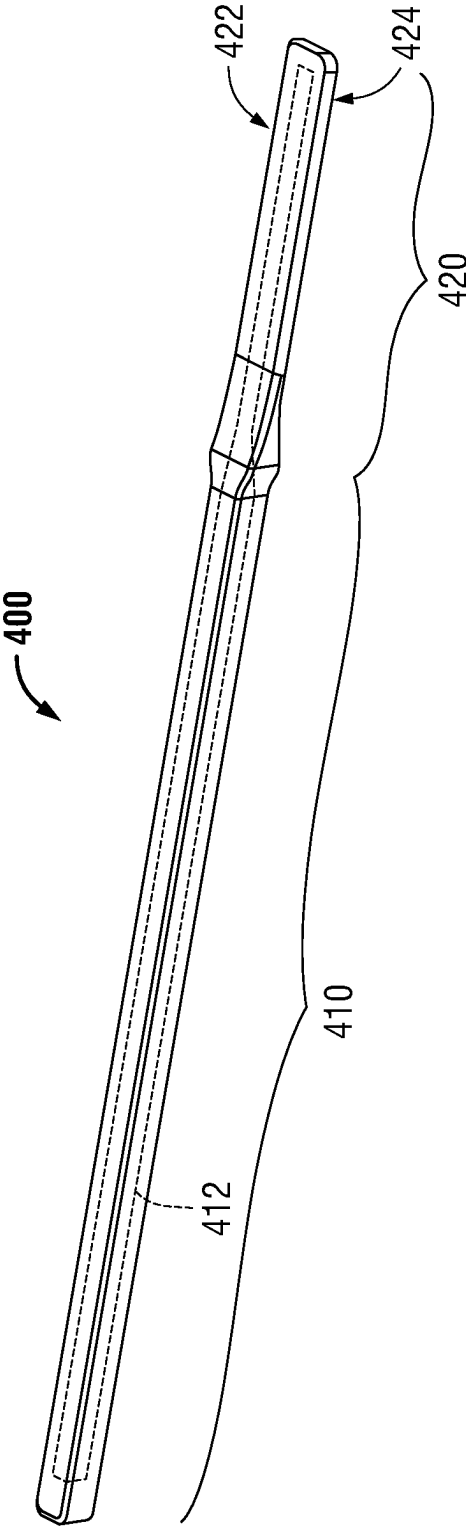


FIG. 4

1

# **ELECTROSURGICAL INSTRUMENT FOR GRASPING, TREATING, AND/OR DIVIDING TISSUE INCORPORATING THERMAL MANAGEMENT FEATURE**

## **CROSS-REFERENCE TO RELATED APPLICATIONS**

This application claims the benefit of, and priority to, U.S. Provisional Patent Application No. 62/900,573, filed on Sep. 15, 2019, the entire contents of which are hereby incorporated by reference herein.

## **FIELD**

The present disclosure relates to electrosurgical instruments and, more particularly, to an electrosurgical instrument for grasping, treating, and/or dividing tissue that incorporates a thermal management feature.

## **BACKGROUND**

A surgical forceps is a plier-like instrument which relies on mechanical action between its jaws to grasp tissue. Electrosurgical forceps utilize both mechanical clamping action and electrical energy to treat tissue, e.g., coagulate, cauterize, and/or seal tissue.

## **SUMMARY**

As used herein, the term “distal” refers to the portion that is described which is further from a user, while the term “proximal” refers to the portion that is being described which is closer to a user. Further, any or all of the aspects described herein, to the extent consistent, may be used in conjunction with any or all of the other aspects described herein.

Provided in accordance with aspects of the present disclosure is an electrosurgical instrument. The electrosurgical instrument includes at least one shaft member. In aspects, the electrosurgical instrument includes first and second shaft members. An end effector assembly extends distally from the at least one shaft member. The end effector assembly includes first and second jaw members, each including an electrically-conductive tissue-contacting surface. At least one of the first or second jaw members is movable relative to the other about a pivot between a spaced-apart position and an approximated position for grasping tissue between the electrically-conductive tissue-contacting surfaces. In aspects where the electrosurgical instrument includes first and second shaft members, the first and second shaft members are movable about the pivot and relative to one another between an open position and a closed position to move the first or second jaw members between the spaced-apart position and the approximated position.

The electrosurgical instrument further includes a heat pipe including a proximal body portion and a flattened distal portion. The proximal body portion extending through at least a portion of the at least one shaft member (the first shaft member, in aspects where first and second shaft members are provided), and the flattened distal portion extends through at least a portion of the first jaw member. The heat pipe is configured to facilitate drawing heat proximally from the first and second jaw members.

In an aspect of the present disclosure, the heat pipe defines an internal lumen extending therethrough. The internal

2

lumen is, in aspects, partially constricted and, in other aspects, fully constricted, in the distal flattened portion of the heat pipe.

In another aspect of the present disclosure, the distal flattened portion of the heat pipe defines a spatula-shaped configuration.

In still another aspect of the present disclosure, the distal flattened portion of the heat pipe includes an upper surface defining a plane disposed in parallel orientation relative to a plane defined by the electrically-conductive tissue-contacting surface of the first jaw member.

In yet another aspect of the present disclosure, the heat pipe includes a cooling fluid disposed therein.

In still yet another aspect of the present disclosure, the at least one shaft member (the first shaft member) includes an inner frame and an outer housing. The proximal body portion of the heat pipe extends through the outer housing and at least one of: alongside, above, below, or through the inner frame.

In another aspect of the present disclosure, a length of the distal flattened portion of the heat pipe is configured in accordance with a length of the electrically-conductive tissue-contacting surface of the first jaw member. Additionally or alternatively, a length of the heat pipe is configured in accordance with a length of the electrically-conductive tissue-contacting surface of the first jaw member. Further still, as an addition or alternative, a cross-sectional area of the heat pipe may be configured in accordance with an exposed surface area of the electrically-conductive tissue-contacting surface of the first jaw member.

## **BRIEF DESCRIPTION OF THE DRAWINGS**

The above and other aspects and features of the present disclosure will become more apparent in light of the following detailed description when taken in conjunction with the accompanying drawings wherein like reference numerals identify similar or identical elements.

FIG. 1A is a perspective view, from a first side, of an electrosurgical instrument provided in accordance with aspects the present disclosure;

FIG. 1B is a side view, from a second, opposite side, of the electrosurgical instrument of FIG. 1A;

FIG. 2A is a perspective view, from the first side, of the electrosurgical instrument of FIG. 1A with portions of the outer housings of the first and second shaft members removed to illustrate the internal components therein;

FIG. 2B is a perspective view, from the second side, of the electrosurgical instrument of FIG. 1A with portions of the outer housings of the first and second shaft members removed to illustrate the internal components therein;

FIG. 3A is a perspective view of the inner frame and jaw member of the first shaft member of the electrosurgical instrument of FIG. 1A;

FIG. 3B is an enlarged, perspective view of the area of detail indicated as “3B” in FIG. 3A; and

FIG. 4 is a perspective view of the heat pipe of the inner frame and jaw member of the first shaft member of the electrosurgical instrument of FIG. 1A.

## **DETAILED DESCRIPTION**

Referring to FIGS. 1A-2B, an electrosurgical instrument provided in accordance with the present disclosure is shown generally identified by reference numeral **100**. Instrument **100** includes first and second shaft members **110**, **120** each having a proximal end portion **112a**, **122a** and a distal end

portion 112*b*, 122*b*. An end effector assembly 200 of instrument 100 includes first and second jaw members 210, 220 extending from distal end portions 112*b*, 122*b* of shaft members 110, 120, respectively. Instrument 100 further includes a pivot 130 pivotably coupling first and second shaft members 110, 120 with one another towards distal end portions 112*b*, 122*b*, respectively, thereof to enable pivoting of shaft members 110, 120 relative to one another between an open position and a closed position to thereby pivot jaw members 210, 220 relative to one another between a spaced-apart position and an approximated position to grasp tissue therebetween. Instrument 100 additionally includes a knife 140, a knife deployment mechanism 150 for selectively deploying knife 140 between jaw members 210, 220 of end effector assembly 200, a knife lockout mechanism 170 for inhibiting deployment of knife 140 prior to sufficient approximation of jaw members 210, 220, and a switch assembly 180 for enabling the selective supply of electro-surgical energy to end effector assembly 200. An electro-surgical cable 300 electrically couples instrument 100 to a source of energy (not shown), e.g., an electrosurgical generator, to enable the supply of electrosurgical energy to jaw members 210, 220 of end effector assembly 200 upon activation of switch assembly 180.

A thermal management feature in the form of a heat pipe 400 (FIGS. 3A-4) extends at least partially through one of the shaft members and the corresponding jaw member, e.g., shaft member 110 and jaw member 210, to facilitate drawing heat proximally away from end effector assembly 200, thus facilitating cooling of end effector assembly 200 after use, minimizing thermal spread during use, and/or reducing harm from accidental contact with end effector assembly 200 during or after use. Heat pipe 400 (FIGS. 3A-4) is described in greater detail below.

Although instrument 100 is illustrated and described herein as a hemostat-style electrosurgical instrument, other electrosurgical instruments such as shaft-based electrosurgical instruments, e.g., electrosurgical instruments including a housing, a shaft extending distally from housing, and end effector assembly 200 disposed at the distal end portion of the shaft; robotic electrosurgical instruments, e.g., electrosurgical instruments including a robotic attachment hub, a shaft extending distally from the attachment hub, and end effector assembly 200 disposed at the distal end portion of the shaft; or other suitable electrosurgical instruments are also contemplated and may include some or all of the features of instrument 100, e.g., heat pipe 400 (FIGS. 3A-4), detailed hereinbelow.

Continuing with reference to FIGS. 1A-2B, each shaft member 110, 120 includes an inner frame 114, 124, an outer housing 116, 126 surrounding at least a portion of the respective inner frame 114, 124, and a handle 118, 128 engaged with the respective outer housing 116, 126 towards the proximal end portion 112*a*, 122*a* of the respective shaft member 110, 120.

Inner frames 114, 124 support jaw members 210, 220, respectively, thereon at the distal ends thereof, provide structural support to shaft members 110, 120, and operably support the internal components disposed within shaft members 110, 120. Further, pivot 130 extends through pivot apertures defined within inner frames 114, 124 to pivotably couple shaft members 110, 120 with one another towards distal end portion 112*b*, 122*b*, respectively, thereof such that, as noted above, pivoting of shaft members 110, 120 relative to one another between the open and closed positions pivots jaw members 210, 220 relative to one another between the spaced-apart and approximated positions to grasp tissue

therebetween. One or both of inner frames 114, 124 may be formed from one or more structures secured to one another. For example, inner frame 114 of shaft member 110 may include a body plate 115*a* and a reinforcing plate 115*b* attached to body plate 115*a*, e.g., via welding, to provide increased lateral stiffness and structural support thereto.

Outer housings 116, 126 of shaft members 110, 120 enclose and/or operably support the internal components disposed within shaft members 110, 120. More specifically, outer housing 116 of shaft member 110 encloses and supports at least a portion of inner frame 114, knife deployment mechanism 150, and knife lockout mechanism 170, while outer housing 126 of shaft member 120 receives electrosurgical cable 300 and encloses and supports at least a portion of inner frame 124, switch assembly 180, and the lead wires 310 of electrosurgical cable 300. Handles 118, 128 are engaged with outer housings 116, 126 towards proximal end portions 112*a*, 122*a* of shaft members 110, 120 and extend outwardly from shaft members 110, 120. Handles 118, 128 define finger holes 119, 129 configured to facilitate grasping and manipulating shaft members 110, 120, e.g., between the open and closed positions.

With additional reference to FIGS. 3A and 3B, end effector assembly 200 of instrument 100, as noted above, includes first and second jaw members 210, 220 supported by and extending distally from respective inner frames 114, 124 of shaft members 110, 120, respectively. Alternatively, in embodiments, one or both of jaw members 210, 220 may be formed as a single piece with its respective shaft member 110, 120. Each jaw member 210, 220 includes an electrically-conductive, tissue-contacting plate 214, 224 and an insulative housing 216, 226 surrounding and supporting the respective tissue-contacting plate 214, 224 thereon. Each jaw member 210, 220 further includes a structural jaw support (not shown) secured to the distal end portions of inner frames 114, 124, of shaft members 110, 120, respectively, and extending distally therefrom. Insulative housings 216, 226 are overmolded or otherwise disposed about the structural jaw supports to enclose jaw members 210, 220 and retain tissue-contacting plates 214, 224, respectively, in position thereon. Each tissue-contacting plate 214, 224 defines a tissue-contacting surface 215*a* (only tissue contacting surface 215*a* of tissue-contacting plate 214 of jaw member 210 is shown) and a longitudinally-extending knife channel 215*b* (only knife channel 215*b* of tissue-contacting plate 214 of jaw member 210 is shown). One or both tissue-contacting plates 214 may include one or more stop members 215*c* (stop members 215*c* are only shown disposed on tissue contacting surface 215*a* of tissue-contacting plate 214 of jaw member 210) associated therewith, e.g., disposed thereon or protruding therethrough, and electrically isolated therefrom, e.g., via forming stop members 215*c* at least partially from an electrically-insulative material and/or an intermediate insulating material between stop members 215*c* and tissue-contacting plate 214. Electrical leads 310 (FIG. 2B) are configured to electrically couple to tissue-contacting plates 214, 224 to enable the supply of electrosurgical energy thereto, e.g., upon activation of switch assembly 180.

Referring back to FIGS. 2A and 2B, knife deployment mechanism 150 is coupled to shaft member 110 and generally includes a pair of opposed triggers 152 extending from either side of shaft member 110, first and second linkages 154, 156, and a biasing spring 158. Linkages 154, 156 operably couple triggers 152 with knife 140 such that actuation of either trigger 152 deploys knife 140 from a retracted position, wherein the cutting edge (not shown) of

5

knife **140** is disposed proximally of jaw members **210**, **220**, to an extended position, wherein knife **140** extends at least partially through knife channels **215b** and between tissue-contacting plates **214**, **224** of jaw members **210**, **220** to cut tissue grasped therebetween. Biasing spring **158** biases knife **140** towards the retracted position.

Knife lockout mechanism **170** works in conjunction with shaft members **110**, **120** to inhibit deployment of knife **140** prior to shaft members **110**, **120** reaching a sufficiently-closed position corresponding to a sufficiently-approximated position of jaw members **210**, **220**. Knife lockout mechanism **170** includes a finger **176** extending from shaft member **110**. With shaft members **110**, **120** disposed in the open position, finger **176** is spaced-apart from outer housing **126** of shaft member **120** such that knife lockout mechanism **170** is maintained in an engaged position engaging knife **140** to inhibit deployment of knife **140**. In order to disengage knife lockout mechanism **170** to disengage knife **140** and permit deployment of knife **140**, shaft members **110**, **120** are sufficiently closed such that a portion of outer housing **126** of shaft member **120** contacts finger **176** of knife lockout **170** and urges finger **176** into housing **116** of shaft member **110** to thereby disengage knife lockout mechanism **170** from knife **140**. With knife lockout mechanism **170** disengaged, knife deployment mechanism **150** may be actuated to deploy knife **140** from the retracted position towards the extended position.

With reference to FIG. 2B, switch assembly **180** is disposed on shaft member **120** and generally includes an activation button **182** and a Printed Circuit Board (PCB) **184**. Activation button **182** is configured to be contacted by the outer housing **116** of shaft member **110** upon sufficient approximation of shaft members **110**, **120** so as to depress activation button **182** and activate switch assembly **180**. PCB **184** supports electrical connectors and contacts that, together with electrosurgical cable **300** (FIG. 1A), electrically couple tissue-contacting plates **214**, **224** of jaw members **210**, **220**, switch assembly **180**, and the energy source (not shown) with one another. As a result, activation of activation button **182** initiates the supply of energy from the energy source (not shown) to jaw members **210**, **220** such that such energy may be conducted through tissue grasped between tissue-contacting plates **214**, **224** of jaw members **210**, **220** to treat, e.g., seal, tissue.

Turning to FIGS. 3A-4, as noted above, heat pipe **400** functions as a thermal management feature to facilitate drawing heat proximally away from end effector assembly **200**. Heat pipe **400**, as also noted above, extends at least partially through shaft member **110** and jaw member **210**. Heat pipe **400**, more specifically, includes a proximal body portion **410** extending at least partially through shaft member **110** and a distal flattened portion **420** extending at least partially through jaw member **210**. Heat pipe **400** is formed from a material with a high thermal conductivity, e.g., copper, polyimide, etc.

Proximal body portion **410** of heat pipe **400** is disposed within outer housing **116** of shaft member **110** and may extend at least partially alongside inner frame **114**, underneath inner frame **114**, above inner frame **114**, within a cut-out defined within inner frame **114**, and/or between plates **115a**, **115b** of inner frame **114**. Proximal body portion **410** of heat pipe **400** may contact at least a portion of inner frame **114** to enable inner frame **114** to serve as a heat sink to dissipate heat from heat pipe **400**. Proximal body portion **410** of heat pipe **400** extends proximally from the distal end of inner frame **114** and may extend, in embodiments, proximally to or beyond the proximal end of inner frame **114**, in

6

other embodiments, proximally at least 75% of the length of inner frame **114**, or, in still other embodiments, proximally at least 50% of the length of inner frame **114**. Additionally or alternatively, proximal body portion **410** may extend proximally through outer housing **116** of shaft member **110**, in embodiments, at least 80% of the length of shaft member **110**, in other embodiments, at least 60% of the length of shaft member **110**, or, in still other embodiments, at least 40% of the length of shaft member **110**. A distal end of proximal body portion **410** may be disposed at the distal end of inner frame **114**, at distal end portion **112b** of shaft member **110**, or at pivot **130**.

Proximal body portion **410** of heat pipe **400** defines an internal lumen **412** enclosed therein and extending longitudinally therethrough. A cooling fluid, e.g., water, saline, etc., is disposed within internal lumen **412** to occupy a portion but not entirely fill internal lumen **412**. In embodiments, the inner wall of proximal body portion **410** of heat pipe **400** that defines internal lumen **412** includes a wick structure (not shown) configured to exert capillary pressure on the cooling fluid when the cooling fluid is in a liquid phase. The wick structure may be a series of grooves or other suitable structure defined on or within the inner wall of proximal body portion **410**.

Distal flattened portion **420** of heat pipe **400** extends distally from proximal body portion **410**, e.g., distally from the distal end of inner frame **114**, distal end portion **112b** of shaft member **110**, or pivot **130**, at least partially through jaw member **210**. More specifically, distal flattened portion **420** extends distally within insulative housing **216** of jaw member **210** along at least a portion of the length of jaw member **210**. Distal flattened portion **420** of heat pipe **400** extends, in embodiments, to the distal end of jaw member **210**, in other embodiments, at least 75% of the length of jaw member **210**, or, in still other embodiments, at least 50% of the length of jaw member **210**.

Distal flattened portion **420** defines a spatula-shaped configuration including opposing upper and lower planar surfaces **422**, **424**, respectively. In other embodiments, upper surface **422** is planar while lower surface **424** is curved, angled, or otherwise configured. Upper surface **422** of distal flattened portion **420** of heat pipe **400** may define a plane disposed in parallel orientation relative to a plane defined by tissue-contacting surface **215a** of tissue-contacting plate **214** of jaw member **210**. Distal flattened portion **420** may extend distally, in embodiments, to or beyond the distal end of tissue-contacting plate **214** of jaw member **210**, in other embodiments, at least 90% of the length of jaw member **210**, in yet other embodiments, at least 80% of the length of tissue-contacting plate **214**, or, in still other embodiments, at least 70% of the length of tissue-contacting plate **214**.

Distal flattened portion **420** may be disposed in direct contact with the underside of tissue-contacting plate **214**, may indirectly contact tissue-contacting plate **214** via an intermediate thermally-conductive structure (in embodiments, a thermally-conductive, electrically-insulative structure), or may be disconnected from tissue-contacting plate **214** but thermally coupled thereto, e.g., via the close proximity of distal flattened portion **420** relative to tissue-contacting plate **214** within insulative housing **216** of jaw member **210**. In embodiments, distal flattened portion **420** is disposed in contact with the structural jaw support of jaw member **210** in thermal communication therewith to absorb heat therefrom. Further, distal flattened portion **420** may at least partially extend underneath knife channel **215b**, along either side of knife channel **215b**, and/or through knife channel **215b** (without interfering with knife **140** (see FIGS.

2A & 2B) during translation of knife **140** through knife channel **215b**). In embodiments, distal flattened portion **420** is bifurcated to surround knife channel **215b** on both sides thereof.

Referring to FIG. 4, heat pipe **400** may be formed as a single body including internal lumen **412** extending there-through and, thereafter, the distal portion of heat pipe **400** may be flattened to define distal flattened portion **420**. Other methods of manufacturing heat pipe **400** are also contemplated. Regardless of the method of manufacturing heat pipe **400**, lumen **412** may extend at least partially through distal flattened portion **420** and may be occluded within distal flattened portion **420** due to the flattened configuration of distal flattened portion **420** to inhibit fluid flow therethrough. Alternatively, lumen **412** may be constricted, e.g., to define a reduced volume, through distal flattened portion **420** as a result of the flattening of distal flattened portion **420** to limit fluid flow therethrough.

With general reference to FIGS. 1A-4, in use, with tissue grasped between tissue-contacting plates **214**, **224** of jaw members **210**, **220**, shaft members **110**, **120** are sufficiently approximated such that shaft member **110** actuates activation button **182** of switch assembly **180** to initiate the supply of electrosurgical energy to tissue-contacting plates **214**, **224**. The electrosurgical energy is conducted through the tissue grasped between tissue-contacting plates **214**, **224** to heat and treat, e.g., seal, the tissue. As the electrosurgical energy is conducted to treat the tissue, jaw members **210**, **220** are heated. As the temperature of jaw member **210** increases, the cooling fluid within internal lumen **412** of heat pipe at distal flattened portion **420** of heat pipe **400** and/or at the distal end portion of proximal body portion **410** of heat pipe **400** absorbs heat from jaw member **210**, thereby transitioning the cooling fluid from a liquid phase to a vapor phase. The cooling fluid, in the vapor phase, travels proximally through heat pipe **500** away from jaw member **210**. Once sufficiently displaced from jaw member **210** and the elevated temperature thereof, the cooling fluid releases the absorbed heat through proximal body portion **410** of heat pipe **400**. As the cooling fluid releases the absorbed heat, the cooling fluid returns from the vapor phase to the liquid phase and thus returns distally towards jaw member **210** to repeat the cycle. In this manner, heat is drawn away from jaw member **210** (and, thus, end effector assembly **200**) and is dissipated within shaft member **110**. In embodiments, it is contemplated that heat pipe **400** cool jaw member **210** down to a temperature of about 60° C. or below, e.g., from 90° C.

In embodiments, heat pipe **400** may define a length relative to a length of tissue-contacting plate **214** of jaw member **210** to facilitate cooling. More specifically, the heat pipe length to tissue-contacting plate length ratio may be, in embodiments, from 1:1 to 2:1; in other embodiments, at least 6:1; in other embodiments, at least 9:1; and in still other embodiments, at least 12:1. In percentage terms, the tissue-contacting plate length may be, in embodiments, from 6% to 12% of the length of the heat pipe; in other embodiments, at least 60% of the length of the heat pipe; and in still other embodiments, at least 80% of the length of the heat pipe.

In embodiments, tissue-contacting plate **214** may define an exposed surface area relative to a maximum, minimum, or average cross-sectional area of heat pipe **400** to facilitate cooling. More specifically, the exposed tissue-contacting plate surface area to heat pipe cross-sectional area ratio may be, in embodiments, from 3:1 to 4:1; in other embodiments, from 20:1 to 35:1; in still other embodiments, from 3:1 to 35:1. In percentage terms, the heat pipe cross-sectional area may be from 2% to 5% of the exposed surface area of the

tissue-contacting plate; in other embodiments, from 20-35% of the exposed surface area of the tissue-contacting plate; and in still other embodiments, from 2-35% of the exposed surface area of the tissue-contacting plate.

From the foregoing and with reference to the various drawings, those skilled in the art will appreciate that certain modifications can be made to the present disclosure without departing from the scope of the same. While several embodiments of the disclosure have been shown in the drawings, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise. Therefore, the above description should not be construed as limiting, but merely as exemplifications of particular embodiments. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

What is claimed is:

1. An electrosurgical instrument, comprising:

at least one shaft member;

an end effector assembly extending distally from the at least one shaft member, the end effector assembly including first and second jaw members, each of the first and second jaw members including an electrically-conductive tissue-contacting surface, at least one of the first or second jaw members movable relative to the other about a pivot between a spaced-apart position and an approximated position for grasping tissue between the electrically-conductive tissue-contacting surfaces; and

a heat pipe including a proximal body portion having a first cross-sectional configuration and a flattened distal portion having a second cross-sectional configuration that is flattened relative to the first cross-sectional configuration, the proximal body portion extending through at least 40% of the at least one shaft member, and the flattened distal portion extending through at least a substantial portion of the first jaw member, the heat pipe defining an internal lumen configured to circulate a cooling fluid between the proximal body portion and the flattened distal portion to facilitate drawing heat proximally from the end effector assembly,

wherein the second cross-sectional configuration constricts the internal lumen in the flattened distal portion of the heat pipe to reduce or inhibit flow of the cooling fluid through the flattened distal portion relative to flow of the cooling fluid through the proximal body portion.

2. The electrosurgical instrument according to claim 1, wherein the distal flattened portion of the heat pipe defines a spatula-shaped configuration.

3. The electrosurgical instrument according to claim 1, wherein the distal flattened portion of the heat pipe includes an upper surface defining a plane disposed in parallel orientation relative to and underneath a plane defined by the electrically-conductive tissue-contacting surface of the first jaw member.

4. The electrosurgical instrument according to claim 1, wherein the at least one shaft member includes an inner frame and an outer housing, and wherein the proximal body portion of the heat pipe extends through the outer housing and at least one of: alongside, above, below, or through the inner frame.

5. The electrosurgical instrument according to claim 1, wherein a length of the distal flattened portion of the heat

9

pipe is configured in accordance with a length of the electrically-conductive tissue-contacting surface of the first jaw member.

6. The electrosurgical instrument according to claim 1, wherein a length of the heat pipe is configured in accordance with a length of the electrically-conductive tissue-contacting surface of the first jaw member.

7. The electrosurgical instrument according to claim 1, wherein a cross-sectional area of the heat pipe is configured in accordance with an exposed surface area of the electrically-conductive tissue-contacting surface of the first jaw member.

8. The electrosurgical instrument according to claim 1, wherein the flattened distal portion of the heat pipe is disposed proximate the electrically-conductive tissue-contacting surface of the end effector assembly.

9. An electrosurgical instrument, comprising:

first and second shaft members movable relative to one another about a pivot;

first and second jaw members extending distally from the first and second shaft members, respectively, each of the first and second jaw members including an electrically-conductive tissue-contacting surface, wherein movement of the first and second shaft members relative to one another from an open position to a closed position moves the first and second jaw members relative to one another from a spaced-apart position to an approximated position for grasping tissue between the electrically-conductive tissue-contacting surfaces; and

a heat pipe including a proximal body portion having a first cross-sectional configuration and a flattened distal portion having a second cross-sectional configuration that is flattened relative to the first cross-sectional configuration, the proximal body portion extending through at least 40% of the first shaft member, and the flattened distal portion extending through at least a substantial portion of the first jaw member, the heat pipe defining an internal lumen configured to circulate a cooling fluid between the proximal body portion and the flattened distal portion to facilitate drawing heat proximally from the first and second jaw members, wherein the second cross-sectional configuration constricts the internal lumen in the flattened distal portion of the heat pipe to reduce or inhibit flow of the cooling fluid through the flattened distal portion relative to flow of the cooling fluid through the proximal body portion.

10. The electrosurgical instrument according to claim 9, wherein the distal flattened portion of the heat pipe defines a spatula-shaped configuration.

11. The electrosurgical instrument according to claim 9, wherein the distal flattened portion of the heat pipe includes an upper surface defining a plane disposed in parallel orientation relative to and underneath a plane defined by the electrically-conductive tissue-contacting surface of the first jaw member.

12. The electrosurgical instrument according to claim 9, wherein the first shaft member includes an inner frame and an outer housing, and wherein the proximal body portion of the heat pipe extends through the outer housing and at least one of: alongside, above, below, or through the inner frame.

13. The electrosurgical instrument according to claim 9, wherein a length of the distal flattened portion of the heat

10

pipe is configured in accordance with a length of the electrically-conductive tissue-contacting surface of the first jaw member.

14. The electrosurgical instrument according to claim 9, wherein a length of the heat pipe is configured in accordance with a length of the electrically-conductive tissue-contacting surface of the first jaw member.

15. The electrosurgical instrument according to claim 9, wherein a cross-sectional area of the heat pipe is configured in accordance with an exposed surface area of the electrically-conductive tissue-contacting surface of the first jaw member.

16. The electrosurgical instrument according to claim 9, wherein the flattened distal portion of the heat pipe is disposed proximate the electrically-conductive tissue-contacting surface of the first jaw member.

17. An electrosurgical instrument, comprising:

at least one shaft member;

an end effector assembly extending distally from the at least one shaft member, the end effector assembly including first and second jaw members, each of the first and second jaw members including an electrically-conductive tissue-contacting surface, at least one of the first or second jaw members movable relative to the other about a pivot between a spaced-apart position and an approximated position for grasping tissue between the electrically-conductive tissue-contacting surfaces; and

a heat pipe including a proximal body portion having a first cross-sectional configuration and a flattened distal portion having a second cross-sectional configuration that is flattened relative to the first cross-sectional configuration, the proximal body portion extending through at least 40% of the at least one shaft member, and the flattened distal portion extending substantially a length of the electrically-conductive tissue contacting surface through the first jaw member, the heat pipe defining an internal lumen configured to circulate a cooling fluid between the proximal body portion and the flattened distal portion to facilitate drawing heat proximally from the end effector assembly,

wherein the second cross-sectional configuration constricts the internal lumen in the flattened distal portion of the heat pipe to reduce or inhibit flow of the cooling fluid through the flattened distal portion relative to flow of the cooling fluid through the proximal body portion.

18. The electrosurgical instrument according to claim 17, wherein the distal flattened portion of the heat pipe defines a spatula-shaped configuration.

19. The electrosurgical instrument according to claim 17, wherein the distal flattened portion of the heat pipe includes an upper surface defining a plane disposed in parallel orientation relative to and underneath a plane defined by the electrically-conductive tissue-contacting surface of the first jaw member.

20. The electrosurgical instrument according to claim 17, wherein the at least one shaft member includes an inner frame and an outer housing, and wherein the proximal body portion of the heat pipe extends through the outer housing and at least one of: alongside, above, below, or through the inner frame.

\* \* \* \* \*